

Catalysts

CATALYSTS SPEED UP REACTIONS BY LOWERING THE ACTIVATION ENERGY REQUIRED.

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Energy

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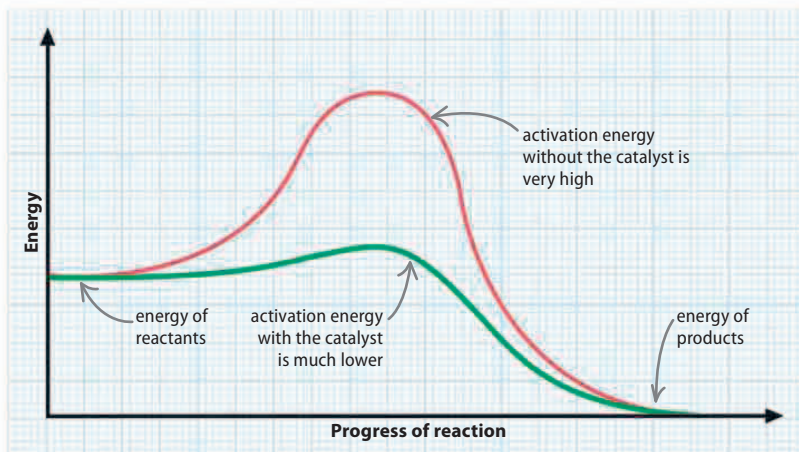
Various catalysts used in laboratories and industry make chemical reactions run faster and allow unreactive materials to get involved in reactions. The enzymes that control reactions in cells are also catalysts.

Less energy needed

Many reactions have activation energies that are so high that the reactions never happen on their own—or happen so slowly that they are hardly noticeable. Catalysts make such reactions possible by reducing the activation energy needed.

► Energy graph

Catalysts reduce the energy barrier between reactants and products. In industry, a catalyst can make reactions more economical.



How catalysts work

Catalysts are a highly varied group of materials. Many are porous substances with tiny spaces inside where the reactants are brought together in such a way that they react without the need for a lot of energy. The precise mechanisms vary but usually a catalyst facilitates a reaction by providing an intermediary phase between the reactants and the products. The catalyst is involved in the reaction but not consumed by it.

Key:



Reactant 1



Reactant 2



Catalyst



Product

► Reactant 1 bonds with catalyst

One reactant bonds temporarily with the catalyst, forming a complex molecule.



► Reactant 2 joins in

The molecule bonds with the other reactant, bringing the reactants close together.



► Product forms

While held in this way, the reactants need much less energy to react. The product can form easily.



► Catalyst breaks away

The product breaks from the catalyst, which is unchanged by the reaction and available to repeat its role.



The word “**catalyst**” comes from the Greek word meaning “to untie.”